

# Digital Twins of Wave Energy Generation Based on Artificial Intelligence

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## ABSTRACT

Ocean waves provide a large amount of renewable energy, and Wave energy converter (WEC) can convert wave energy into electric energy. This paper proposes a visualization platform for wave power generation. The platform can monitor various indicators of wave power generation in real time, combined with Long Short-Term Memory (LSTM) neural network to predict wave power and electricity consumption. We make digital twins of a wave power plant in a computer, allowing users to remotely view the factory through VR glasses.

**Keywords:** Wave Power, Digital Twins, Visualization, AI.

**Index Terms:** Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

## 1 INTRODUCTION

As the environmental problems caused by the use of fossil energy become more and more serious, people are looking for more clean energy to replace the current fossil energy. The energy contained in the ocean is gradually being valued by people, and wave energy is considered to be a high-quality energy because of its wide distribution of energy, relatively simple utilization, and less loss in the process of energy transmission. According to data [1], the power generation rate of WEC is as high as 57%, while the power generation rate of wind and solar power generation devices is 20%-30%. According to the statistics of UNESCO [5], the available energy in the ocean covering 70% of the world is about  $7.6 \times 10^8$  kW, of which the actual exploitable amount of wave energy and salinity difference energy is the largest, which is about  $3 \times 10^8$  kW, and in the actual development process, the development of wave energy accounts for the largest proportion, accounting for about 40%.

Since the marine environment is affected by various factors such as light, temperature, wind speed, etc. [2], the energy output power of the wave energy power generation method has the problem of large fluctuation, which causes a lot of inconvenience for the integration into the power grid, and more accurate power prediction is an essential link in the future development of wave energy. Power forecasting can provide basic information for real-time power forecasting of wave power generation, automatic dispatching of power generation equipment, operation of wave power stations, etc. A mature power forecasting scheme will make wave power generation more competitive in the market. Therefore, we make digital twins system to increase interactivity of power generation monitoring on the basis of visualization. By monitoring various real-time data and applying mature neural network prediction algorithms to simulate the trend of wave motion, it can provide wave power generation

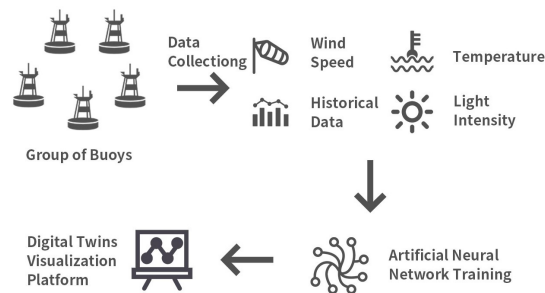


Figure 1: The Overall Process of The System

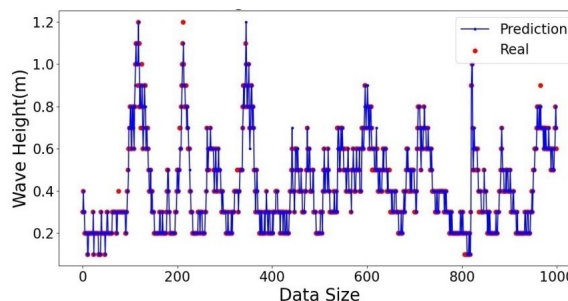


Figure 2: Wave Height Prediction Results

power prediction data and further improve the utilization efficiency of ocean wave energy.

## 2 RESEARCH METHODS

The main objective of our work is to predict the power generation of wave energy equipment. The overall process is to monitor the key data such as temperature, light intensity, wind speed, wave height, etc. [3] that affect the future wave height in real time through the Internet of Things terminal, and then process these data through the AI algorithm model to predict the wave height in the future, and calculate the power generation based on the power generation efficiency of the WEC. Finally, the system output power generation and other IoT monitoring data to the digital twins visualization platform.

### 2.1 AI Power Forecast

In this paper, the Long Short-Term Memory (LSTM) model is used to make a preliminary prediction of ocean wave height, and then use the obtained wave height data as a training set to predict the power generation. LSTM is suitable for processing and predicting events with very long intervals and delays in time series. We compared and tested a variety of popular artificial neural network models. From the comparison of our prediction results on public data and the actual value, the LSTM network can predict the wave height in a short time as shown in Figure 2.

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Figure 3: Power Generation Process Model

Table 1: The Function Module in Energy Monitoring

| Module                 | Features                                                                         |
|------------------------|----------------------------------------------------------------------------------|
| Energy Consumption     | Shows the energy consumed by the system's current device operation.              |
| Running Record         | Records the operating records of the system for the past seven days.             |
| Equipment Efficiency   | Monitor the efficiency of the equipment in real time.                            |
| Load condition         | Real-time monitoring of the load of the system to avoid overload.                |
| Energy conversion rate | Display the current energy conversion rate of the system.                        |
| Equipment status       | Real-time monitoring of the operating status of each power generation equipment. |
| Network transmission   | Monitor the network status of the equipment and discover network abnormalities.  |

## 2.2 Digital Twins

At present, there is no unified definition of digital twins. According to the current investigation [4], digital twins represents a fine modeling method, the technical focus is data collection, and the main feature is that the interaction has the effect of simultaneous interpretation. The application of digital twins in the industrial has some successful cases, which can significantly improve the efficiency and intelligence of industrial manufacturing.

## 3 SOLUTION

### 3.1 Digital Twins Visualization Platform

We display the data predicted by the artificial neural network and the monitoring data in our digital twins visualization platform. From the continuous monitoring of wave power generation equipment, to the real-time monitoring of various power generation data, and the data feedback on the demand side of the distribution network, as shown in Figure 4 and Table 1, our solutions can realize the visualization of the wave power generation system, which can monitor the entire process of energy flow. In addition, more types of data can be showed on other undisplayed visualization pages, which are listed in the first three items in Table 1. The real-time data of nodes provides data support for power grid operations and improves overall operational efficiency. From the interface in the figure, you can see the digital twins power plant model we created, and you can see the real-time data display of each monitoring module on the background of the model.

### 3.2 Remote Control

While designing the visualization platform, we also considered another important feature of digital twins application which is simultaneous interpretation and interaction. Simultaneous interpreting

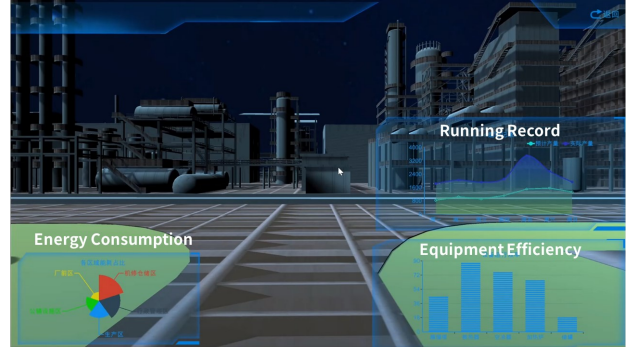


Figure 4: View Wave Power Generation Center in Real Time

interaction means that users can interact with terminal devices without physically present. We use VR equipment combined with the scene created by Unity 3D to operate the equipment projection in the virtual scene, and detect the impact of different operations on the overall power generation efficiency through various feedbacks of the system, so as to make optimal decisions in the power generation process.

In the next step, we hope to realize real simultaneous interpretation interaction, that is, the user's operations in the digital twins can operate the physical entity in real time through the IoT terminal.

## 4 CONCLUSION

As you can see in our solution, our digital twins visualization platform allows users to intuitively see the operating status and various operating data of wave power generation equipment. We use artificial neural network algorithm to establish a data prediction model of wave height and power generation, select the algorithm with the highest prediction accuracy to build the model, and guide the staff to reasonably arrange the power generation plan for a period of time in the future according to the prediction results, which provides a reliable guarantee to the stable operation of the power generation process. Digital twins can solve many problems that currently exist in the wave power generation process, strengthen the correlation of various data, and facilitate the decision-making of staff.

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